



Cambridge International AS & A Level

PHYSICS

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Paper 4 A Level Structured Questions

October/November 2020

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

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ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
MO	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

Question	Answer	Marks
1(a)	work done per unit mass	B1
	(work done) moving mass from infinity (to the point)	B1
1(b)(i)	gravitational potential energy = $(-)\frac{GMm}{r}$	C1
	$\Delta E_p = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 2.4 \times 10^3 \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	C1
	or	
	$\Delta \phi = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times [(6.4 \times 10^6)^{-1} - (1.2 \times 10^7)^{-1}]$	(C1)
	$\Delta E_p = m\Delta \phi$	(C1)
	$\Delta E_p = 7.0 \times 10^{10} \text{ J}$	A1
1(b)(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$	C1
	$v^2 = \frac{GM}{r}$ $= (6.67 \times 10^{-11} \times 6.0 \times 10^{24}) / (1.2 \times 10^7)$	C1
	$v = 5800 \text{ m s}^{-1}$	A1
1(c)	any one point from: - smaller gain in energy required if orbit is west to east - smaller change in velocity if orbit is west to east - smaller gain in energy if orbit is in same direction as Earth's rotation - smaller change in velocity if orbit is in same direction as Earth's rotation - satellite already moving west to east at launch - Earth's rotation is from west to east	B1

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Question	Answer	Marks
2(a)	sum of potential energy and kinetic energy (of particles)	B1
	(total) energy of random motion of particles	B1
2(b)(i)	$pV = nRT$	C1
	$2.60 \times 10^5 \times 2.30 \times 10^{-3} = n \times 8.31 \times 180$	A1
	$n = 0.400 \text{ mol}$	
2(b)(ii)	$(2.30 \times 10^{-3}) / 180 = (3.80 \times 10^{-3}) / T$	C1
	or $2.60 \times 10^5 \times 3.80 \times 10^{-3} = 0.400 \times 8.31 \times T$	
	$T = 297 \text{ K}$	A1
2(c)(i)	$\Delta W = p\Delta V$	C1
	$= 2.60 \times 10^5 \times (2.30 - 3.80) \times 10^{-3}$	
	$= (-)390 \text{ J}$	A1
	negative because work is done by gas or negative because work is done against atmospheric pressure or negative because volume of gas increases	B1
2(c)(ii)	$\Delta U = (980 - 390)$ $= 590 \text{ J}$	A1

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Question	Answer	Marks
3(a)	acceleration (directly) proportional to displacement	B1
	acceleration is in opposite <u>direction</u> to displacement or acceleration is (directed) towards a fixed point	B1
3(b)(i)	zero	B1
3(b)(ii)	E_T is maximum potential energy = mgh $E_T = 94 \times 10^{-3} \times 9.81 \times 0.90 \times 10^{-2}$	C1
	$= 8.3 \times 10^{-3} \text{ J}$	A1
3(b)(iii)	$E_{\text{MAX}} = \frac{1}{2} mv_0^2$ and $v_0 = \omega x_0$ or $E_{\text{MAX}} = \frac{1}{2} m(\omega x_0)^2$	C1
	$8.3 \times 10^{-3} = \frac{1}{2} \times 94 \times 10^{-3} \times \omega^2 \times (12.7 \times 10^{-2})^2$...leading to $\omega = 3.3 \text{ rad s}^{-1}$	A1
3(c)	$T = 2\pi / \omega$	C1
	$2\pi / 3.3 = 2\pi \times (L / 9.81)^{1/2}$	C1
	$L = 0.90 \text{ m}$	A1

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Question	Answer	Marks
4(a)	any two points from: • signal can be regenerated/noise can be removed • signal can be encrypted • signal can be checked for errors • multiplexing is possible • <u>circuits</u> are more reliable/cheaper • <u>data</u> can be transmitted at a greater <u>rate</u>	B2
4(b)(i)	right-hand zero underlined (011 <u>0</u>)	B1
4(b)(ii)	analogue signals given as: 3.0, 4.8, 1.0	B1
	0011 at 0.30 ms and 0001 at 0.50 ms	B1
	0100 at 0.40 ms	B1
4(c)	series of steps, all of width 0.1 ms	B1
	steps levels, in order, at output voltage 0, 5, 6, 3 and 4 mV 2 marks: all levels correct 1 mark: one level incorrect and all others correct or one level omitted and last step shown at 1 mV	B2

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Question	Answer	Marks
5(a)(i)	region (of space)	B1
	where a particle experiences a force	B1
5(a)(ii)	similarity – any one point from: • both have an inverse square variation • both decrease with distance • both are radial	B1
	difference – any one point from: • gravitational field always towards (the mass) • electric field can be towards or away from (the charge)	B1
5(b)(i)	$E = Q / 4\pi\epsilon_0 x^2$	C1
	$Q = 4\pi \times 8.85 \times 10^{-12} \times 84 \times 0.15^2$ $= 2.1 \times 10^{-10} \text{ C}$	A1
5(b)(ii)	$E = 84 \times (0.15 / 0.45)^2$ or $E = (2.1 \times 10^{-10}) / (4\pi \times 8.85 \times 10^{-12} \times 0.45^2)$	C1
	$E = 9.3 \text{ V m}^{-1}$	A1
5(c)	line at $E = 0$ from $x = 0$ to $x = 0.15 \text{ m}$	B1
	smooth curve with decreasing negative gradient throughout, from $x = 0.15 \text{ m}$ to $x = 0.45 \text{ m}$, passing through (0.15, 84)	B1
	line passing through (0.45, 9.3)	B1

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Question	Answer	Marks
6(a)(i)	charge per unit potential (difference)	M1
	charge on one plate and potential difference between the plates	A1
6(a)(ii)	any three points from: - smoothing - timing/(time) delaying - tuning - oscillator - blocking d.c. - surge protection - temporary power supply	B3
6(b)(i)	parallel combination of two in series and a single capacitor	B1
6(b)(ii)	one capacitor in series with two in parallel	B1

Question	Answer	Marks
7(a)	X-ray photon produced when electron is decelerated	B1
	larger acceleration results in larger photon energy	B1
	continuous range of accelerations so continuous spectrum of wavelengths/frequencies	B1
7(b)	electron in (inner shell of) target atom is excited (on collision)	B1
	electron de-excites causing emission of a photon	B1
	discrete energy levels so discrete photon wavelengths	B1

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Question	Answer	Marks
8(a)(i)	gain is the same for all frequencies	B1
8(a)(ii)	no (time) delay in change in output when input is changed	B1
8(b)(i)	(at saturation,) $V_{OUT} = 5.0 \text{ V}$	C1
	gain = $5.0 / 0.40$ = 12.5 or 13	A1
8(b)(ii)	$12.5 = 1 + (R / 800)$	C1
	$R = 9200 \Omega$	A1

Question	Answer	Marks
9(a)(i)	(induced) e.m.f. (directly) proportional to rate	M1
	of change of magnetic flux (linkage)	A1
9(a)(ii)	e.m.f. = 0 apart from thin pulses at t_1 and t_2	B1
	rectangular pulses centred on t_1 and t_2 , of widths 2 small squares and 1 small square respectively	B1
	e.m.fs. at t_1 and t_2 have opposite polarities	B1
	magnitude of e.m.f. at t_2 double the magnitude of e.m.f. at t_1	B1
9(b)	V_H shown as zero before ($t_1 - 2$ squares) and after ($t_2 + 2$ squares) and rises to a constant non-zero value between t_1 and t_2	M1
	change at t_1 shown as 2 small squares wide and change at t_2 shown as 1 small square wide	A1

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Question	Answer	Marks
10(a)	concentric circles centred on the wire	B1
	separation of lines increasing with distance from wire	B1
	arrows show anti-clockwise direction	B1
10(b)(i)	current in (each) wire creates a magnetic field (at the other wire)	B1
	current (in wire) at 90° to field causes force	B1
10(b)(ii)	force on each wire towards other wire/attractive	B1
10(c)	Newton's third law pair of forces so yes (forces are equal) or force proportional to product of both currents so yes (forces are equal)	B1

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Question	Answer	Marks
11(a)	any two points from: (maximum) kinetic energy of electrons is independent of intensity - maximum kinetic energy of electrons depends on frequency - no time delay (between illumination and emission)	B2
11(b)(i)	(for $E_{\text{MAX}} = 0$), $1/\lambda_0 = 1.93 \times 10^6 \text{ (m}^{-1}\text{)}$	C1
	$f_0 = 3.00 \times 10^8 \times 1.93 \times 10^6$ $= 5.8 \times 10^{14} \text{ Hz}$	A1
11(b)(ii)	$hc/\lambda = \phi + E_{\text{MAX}}$	C1
	$hc = \text{gradient}$	C1
	gradient = e.g. $[(0.40 - 0.20) \times 1.60 \times 10^{-19}] / [(2.25 - 2.09) \times 10^6] \text{ (working needed)}$ $(= 2.0 \times 10^{-25})$	M1
	$h = (2.0 \times 10^{-25}) / (3.00 \times 10^8) = 6.7 \times 10^{-34} \text{ J s (both working and answer needed)}$	A1
11(c)	straight line with same gradient as the original	B1
	straight line with x-axis intercept greater than $1.93 \times 10^6 \text{ m}^{-1}$	B1

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Question	Answer	Marks
12(a)(i)	energy required to separate nucleons (of nucleus)	M1
	to infinity	A1
12(a)(ii)	a (single) large nucleus <u>divides</u> to form (smaller) nuclei	B1
	any one point from: - initiated by neutron bombardment - resulting nuclei are of similar size - binding energy per nucleon increases - total binding energy increases - neutrons released - combined mass of smaller nuclei is less than mass of large nucleus	B1
12(b)	binding energy per nucleon is a maximum at around $A = 56$	B1
	products of splitting a ^{56}Fe nucleus must have a lower total binding energy	B1
	(reaction would require) a net input of energy	B1